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AGE 418
Sensors and Controls in Precision Agriculture

Lecture 9 Compendium:
Yield Monitoring and Mapping

The "Report Card" of the Farm

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1 1. Introduction to Yield Monitoring

Yield Monitoring is the process of measuring crop yield in real-time as the harvester moves through the field. It creates a dataset that links mass flow (yield) to specific GPS coordinates.

1.1 Why is it the "First Step"?

For many farmers, a Yield Map is their entry point into Precision Agriculture because:

- **Visual Proof:** It visually proves that the field is not uniform. Seeing red spots (low yield) next to green spots (high yield) quantifies the variability.
- **Economic Baseline:** It tells the farmer exactly which parts of the field are profitable and which are losing money.
- **Feedback Loop:** It acts as the "Report Card" for the season. Did the variable rate fertilizer plan work? The yield map provides the answer.

2 2. System Architecture

A yield monitor is not a single sensor; it is a system of synchronized components working together.

2.1 Core Components

1. **Mass Flow Sensor:** Measures the weight or volume of the grain stream.
2. **Moisture Sensor:** Measures water content to calculate dry weight.
3. **GNSS Receiver:** Records location (Latitude, Longitude, Elevation).
4. **Header Position Sensor:** Tells the computer when to record (Harvesting) vs. ignore (Turning).
5. **Display/Console:** The user interface in the cab that visualizes the map.

3 3. Mass Flow Sensors

How do we weigh grain moving at 100 tons per hour without stopping?

3.1 1. Impact Plate (The Industry Standard)

This type is used in most combines (e.g., John Deere, Case IH).

- **Location:** Positioned at the very top of the Clean Grain Elevator.
- **Mechanism:** As grain is thrown off the elevator paddles, it hits a plate.
- **Physics:** The force of the impact is measured by a Load Cell (strain gauge).
- **Formula:** Force = Mass $\times$ Acceleration. Since the speed of the elevator (acceleration) is constant, the Force is directly proportional to the Mass.

3.2 2. Optical Sensors (Volumetric)

Used in some European machines (e.g., CLAAS).

- **Location:** A light beam across the elevator.
- **Mechanism:** Emitter sends light; Receiver measures how much light is blocked by the grain.
- **Calculation:** Measures Volume. To get Mass, the system must know the grain density (Test Weight).
- **Con:** Less accurate if grain density varies significantly across the field.

4 4. Moisture Sensors

Farmers sell "Dry Grain." If you harvest maize at 18% moisture but the market standard is 15.5%, you are selling water weight. To get the **True Yield**, the monitor must subtract this water.

4.1 Capacitance Sensor

- **Location:** Usually mounted on the side of the clean grain elevator. A small auger diverts a sample of grain past the sensor plate.
- **Principle:** Dielectric constant (as discussed in Lecture 4). Wet grain conducts electricity differently than dry grain.
- **Challenge - Sensor Buildup:** Sap, dust, and green material can coat the sensor plate, causing false high-moisture readings. Frequent cleaning is required.

5 5. Data Processing: The "Lag Time"

One of the most critical engineering concepts in yield monitoring is **Flow Delay** or **Lag Time**.

5.1 The Problem

When the header cuts the crop at Point A, it takes time (e.g., 12 seconds) for that grain to travel through the feeder house, thresher, sieves, and elevator to hit the sensor at Point B.

- If the GPS records location at Time $T = 0$...
- But the Sensor records Mass at Time $T = 12$...
- The map will be shifted forward by 12 seconds of travel distance (roughly 20-30 meters).

5.2 The Solution

The software must apply a **Negative Time Shift** to align the Mass data with the correct GPS coordinate. This is usually calibrated by the operator.

- **Too Short Lag:** Yield appears before the harvester enters the crop.
- **Too Long Lag:** Yield appears to continue after the harvester leaves the crop.

6 6. Calibration

A yield monitor is only an estimator. It must be calibrated against a known weight to be accurate.

6.1 Calibration Procedure

1. Harvest a full tank (or a specific calibration load, e.g., 3000 kg).
2. Weigh the load on a static scale (Weighbridge or Grain Cart scale).
3. Enter the "Actual Weight" into the monitor.
4. The monitor calculates a **Calibration Coefficient** to correct its curve.

Best Practice: Calibrate for different flow rates (Low speed, Medium speed, High speed) to ensure linearity across different yield potentials.

7 7. Interpreting Yield Maps

Once the map is generated, it shows spatial variability. We interpret this to make management decisions.

7.1 Identifying Patterns

- **Straight Lines / Strips:** Usually Man-Made issues.
  - *Example:* A clogged sprayer nozzle earlier in the season, planter malfunctions, or compaction from tire tracks.
- **Irregular / Blobs:** Usually Natural issues.
  - *Example:* Soil type changes (sand lens), topography (low spot drowning), or pest infestation patches.

7.2 Normalized Yield

To compare different crops (e.g., Maize 2023 vs Beans 2024), we convert yield to a percentage of the field average.

- **Stable High Zones:** Always > 120% of average. (Good Soil).
- **Stable Low Zones:** Always < 80% of average. (Bad Soil).
- **Unstable Zones:** Fluctuates year to year (Weather dependent).

8 8. Lecture Assessment

8.1 Part A: Standard Concept Questions

1. Define **Yield Monitoring**. Why is it considered a "reactive" rather than "proactive" technology?
2. List the five main hardware components of a yield monitoring system.
3. Explain the working principle of an **Impact Plate** mass flow sensor. What physics formula governs its operation?
4. Why is moisture measurement critical for calculating "Dry Yield"?
5. What is **Lag Time** (Flow Delay)? If lag time is set incorrectly (too short), how will the yield map look near the edges of the field?
6. Why is the **Header Position Sensor** necessary for accurate data collection?
7. Describe the calibration process for a yield monitor. Why must it be done for different crop types?
8. Explain the difference between **Man-Made Variability** and **Natural Variability** on a yield map. Give one visual example of each.
9. What is **Normalized Yield**? Why do we use percentages instead of tons/ha when comparing multi-year data?
10. How does a Yield Map serve as the basis for next year's **Variable Rate Fertilizer** plan?

8.2 Part B: Advanced Analysis (Data & Logic)

1. **Calibration Calculation:** A combine monitor reports a harvested load of 3,500 kg. The actual weight measured at the scale is 3,850 kg. *Task:* Calculate the **Error Percentage**. If the current calibration factor is 1.05, what should the new calibration factor be to correct this error?
2. **Lag Time Geometry:** A harvester travels at 6 km/h. The lag time (delay) from cutter bar to sensor is 15 seconds. *Task:* Calculate the distance (in meters) the machine travels during this delay. If this shift is not corrected, how far "off" will the yield map be from the soil map?
3. **Data Cleaning Logic (Pseudocode):** Raw yield data contains errors (e.g., when the combine turns). *Task:* Write a simple logic statement to filter out bad data points. *Variables:* HeaderStatus (0=Up, 1=Down), Speed (km/h), Yield (t/ha). *Condition:* We only want data where the header is down, speed is reasonable (> 2 km/h), and yield is realistic (< 20 t/ha).
4. **Satellite Yield Proxy:** In Tanzania, many smallholders harvest manually (no yield monitor). *Task:* Explain how **High-Resolution Satellite Imagery** (e.g., Planet or Sentinel-2) taken 2 weeks before harvest can be used as a "Proxy" (Substitute) for a physical yield map. What vegetation index correlates best with biomass/yield at that stage?

5. **Sugar Cane Yield Monitoring:** Sugar cane is unique; it is a massive stalk crop, not a grain. *Task:* Research how yield monitoring is done on a **Sugar Cane Chopper Harvester** (e.g., Case IH Austoft). It does not use an impact plate. What mechanism (Weigh Cells / Hydraulic Pressure) is used instead?